

MyTradingPal

Human-Computer Interaction 2025-2026

Phase 3 - Hi-fi prototype and user evaluation

3LEIC03 G05

José Barbosa Reis (up20207304@up.pt)

Mário Landim Moraes Marques Pereira (up202304965@up.pt)

Miguel Tomás Vieira Rodrigues (up202205749@up.pt)

Vasco António Liberal Rodrigues Queirós de Sá (up202306731@up.pt)

Porto, 18 December 2025

Contents

Introduction	4
Part I - User and Task Analysis	4
Project's idea description	4
Related apps/services/systems	5
Questionnaire highlights	5
PACT Analysis	6
People	6
Activities	6
Contexts	7
Technologies	7
Personas	8
Persona 1	8
Persona 2	8
Activity scenarios	9
Activity scenario 1	9
Activity scenario 2	10
Functionalities	10
Core functionalities	10
Supportive functionalities	11
Part II - Lo-fi prototype and heuristic evaluation	12
Project abridged description	12
Prototype's Wireflow	12
Check Portfolio Performance	12
Configure & Launch AI Character	13
Analyze AI Recommendation & Execute Trade	14
Heuristic Evaluation Results	14
Problem #1 - Inefficient Interaction Flow	14
Problem #2 - Lack of Risk Transparency	15
Problem #3 - Inefficient Persona Comparison	16
Problem #4 - Missing Help and Documentation	17
Corrections to perform in Phase 3	17
Solution #1 - Implement side-by-side comparison interface	17
Solution #2 - Enhance risk score transparency	18
Solution #3 - Develop comprehensive help system	18

Part III - Hi-fi prototype and user evaluation	19
Prototype's Wireflow	19
Check Portfolio Performance	19
Configure & Launch AI Character	20
Analyze AI Recommendation & Execute Trade	21
User evaluation protocol	21
Objective	22
Target Users	22
Methodology	22
Pre-test	22
Pre-test Questionnaire	23
Measures	23
Task Execution	23
Post-test Questionnaire	24
Conclusion	24
Results	24
Sample characterization	24
Results and statistical analysis of task 1	25
Results and statistical analysis of task 2	26
Results and statistical analysis of task 3	27
SUS Scores	28
Discussion	29
Conclusion	30
Annexes	31
Phase 1 Annexes	31
Questionnaire	31
Results	31
Phase 2 Annexes	31
Sent Heuristic Evaluation Reports	31
Received Heuristic Evaluation Reports	31
Phase 3 Annexes	31
Evaluation Protocol	31
Evaluation Results	31
Prototype	31

Introduction

MyTradingPal is a mobile-first investment application designed to bridge the gap between financial anxiety and active portfolio management for young adults. Targeted primarily at users aged 18 to 30 who lack financial literacy, the project aims to create a safe, educational environment where users can invest real money or practice in simulation mode, guided by AI characters with distinct trading personalities. By combining automated portfolio management with plain-language explanations, the application seeks to empower beginners (like our primary persona, Sofia) to overcome their fear of complex financial markets.

The project has evolved through a structured design process starting with Phase 1, which validated that the core problem for this demographic was not a lack of interest, but rather a lack of a "safe environment" to learn. This foundation informed the development of the initial interface in Phase 2, which was subsequently refined using Nielsen's Heuristic Evaluation. Critical improvements were implemented during that phase, such as enhancing the visibility of risk scores and introducing side-by-side comparison tools to better support decision-making.

This report details Phase 3 of the project: the empirical validation of the High-Fidelity prototype through user evaluation. We conducted usability testing with 16 participants, matching our target demographic of university students and recent graduates, to assess the system's performance in real-world scenarios. The evaluation focused on three distinct tasks of increasing complexity: checking portfolio performance, configuring an AI persona, and analyzing a trade recommendation. The following sections present the quantitative data and qualitative feedback gathered during these sessions, analyzing whether MyTradingPal has successfully achieved its goal of making investment accessible, transparent, and less intimidating for the next generation of investors.

Part I - User and Task Analysis

Project's idea description

MyTradingPal is a mobile-first investment application that combines AI-driven portfolio management with interactive, character-based learning.

Users can choose to invest real money or start in a risk-free simulation mode, guided by AI characters that represent distinct trading personalities such as conservative, balanced, or aggressive. Each character provides plain-language explanations for its investment decisions, helping users understand not just what is happening, but why.

Designed for a generation of young, mobile, and curious investors, the app makes investing accessible, engaging, and transparent. It features real-time performance monitoring, explainable AI recommendations, and customizable risk settings.

Strong emphasis is placed on trust, privacy, and education through biometric authentication and privacy modes for public use.

Ultimately, the platform empowers users to learn, experiment, and grow their wealth confidently, bridging the gap between financial literacy and real-world investing through an AI experience that feels both personal and intelligent.

Related apps/services/systems

Several existing applications and platforms served as inspiration and reference points for our project. Among the most relevant are eToro, Robinhood and Trading212, each offering features that align with certain aspects of our concept while also revealing gaps our solution aims to fill.

eToro is a popular social trading platform that allows users to invest in real assets and copy the portfolios of professional traders. It demonstrates the appeal of community-driven investing and the power of accessible trading tools. However, its focus on imitation over understanding limits its educational value, providing little insight into why investment decisions are made.

Robinhood, known for commission-free trading and a simple mobile interface, set the standard for accessibility and real-time trading experiences. While it successfully opened financial markets to a broader audience, it lacks transparency and educational support, often encouraging impulsive behavior rather than informed decision-making.

Trading212 combines real-money trading with a demo mode for practice, making it an important reference for our own dual-mode design. It appeals to both beginners and experienced investors but offers limited personalization and no character-based interaction, resulting in a more transactional experience than an engaging learning one.

Our project differentiates by combining the strengths of these systems: real money investing, educational guidance, and automation into a single platform. By introducing AI characters with distinct personalities, we humanize the investing process, turning financial learning into an interactive and personalized experience. Unlike existing tools, our platform emphasizes transparency, trust, and engagement, explaining every AI decision in plain language while allowing users to experiment, learn, and invest confidently in real markets or safe simulations.

Questionnaire highlights

The user questionnaire provided valuable insights into the target audience's demographics, behaviors, expectations, and pain points, guiding both design and technical priorities of the project.

- 81% aged 18–24, 71% students: Young, learning-focused audience familiar with gamified and social apps.
- 45% beginners: Low financial literacy; require plain language and guided learning.

- 84% mobile users: Mobile-first design with one-handed, touch-friendly interaction is essential.
- 28% skeptical of automation: Explainable, transparent AI needed to build trust.
- 87% expect daily updates: Real-time notifications and dynamic feedback drive engagement.
- High privacy concern: Discreet viewing modes and biometric security required.

Overall Takeaway:

Users are young, curious, and mobile.

They want to learn safely, invest confidently, and trust AI characters that explain decisions clearly and accessibly.

PACT Analysis

People

The app targets users aged 18–45, mainly students (71%) aged 18–24, with varying investment experience — 45% beginners, 36% intermediate, and 19% experienced. Users are mobile-first and accustomed to gamified, social interfaces.

Cognitively, they face challenges understanding financial terminology and require visual, simplified explanations. 28% are skeptical of automated systems, making transparency and control vital.

Physically, the app must support one-handed, touch-based use with large, accessible elements. Socially, motivations range from learning to profit and peer comparison, but privacy remains critical.

Design Implications:

- Implement plain language throughout with contextual tooltips for unavoidable financial terminology
- Create progressive onboarding with gamified first-run tutorials that build confidence
- Establish clear visual and functional separation between simulation and real-money modes
- Provide user-adjustable UI complexity levels to accommodate varying expertise
- Ensure comprehensive accessibility: scalable fonts and high-contrast modes
- Build trust through transparent AI explanations and user control primacy
- Design for one-handed mobile operation

Activities

Users perform five main actions: monitor portfolios, customize AI characters, configure AI portfolios, run simulations, and review recommendations.

Usage is frequent: 58% check multiple times weekly, 87% expect daily updates. Sessions range from 30-second glances to 15-minute analyses.

Tasks vary in complexity, requiring glanceable dashboards, guided workflows, and confirmations for high-risk actions. Social engagement is optional but supports learning and motivation.

Design Implications:

- < 3 seconds dashboard comprehension with clear visualizations.
- Explainable AI summaries with expandable details.
- Sandbox mode for risk-free simulation.
- Explicit confirmations and decision history logs.
- Smart, non-intrusive notifications.
- Optional community features with privacy controls.

Contexts

Users access the app mostly on mobile (84%), in public or semi-private spaces, requiring discreet design. Connectivity can be unstable, so offline caching and data sync are essential.

The student demographic expects integrated tutorials, learning progression, and safe experimentation.

As a regulated real-money app, compliance with GDPR, audit trails, and financial disclaimers is mandatory.

Design Implications:

- Adaptive mobile layouts and lighting modes.
- Offline viewing with freshness indicators.
- Efficient caching, compression, and synchronization.
- Privacy mode and minimal notification previews.
- Integrated learning features and compliance workflows.
- Cross-device sync and continuity.

Technologies

The system combines real-time APIs, an AI/ML investment engine, and secure databases within a microservices architecture.

- **Frontend:** React Native apps + responsive PWA.
- **Integrations:** Broker APIs, payment gateways, biometric login (90%), and push notifications.
- **Security:** AES-256 encryption, 2FA, and session management ensure user protection.
- **Accessibility:** WCAG 2.1 AA compliance.

Design Implications:

- < 2 seconds load times, optimized charts, and lazy loading.

- End-to-end encryption and audit logging.
- Real-time WebSocket data with offline fallback.
- Accessible, scalable, and performance-monitored system.
- A/B testing and analytics for continuous improvement.

Personas

Persona 1

Name: Sofia Mendes

Quote: “I want to learn how to invest without feeling overwhelmed.”

Age: 25 years

Education level: Bachelor’s degree in Business Administration

Work/occupation: Junior Marketing Analyst

Family: Single, no children

Location: Porto, Portugal

Tech proficiency: Moderate — comfortable using apps and social media, but not experienced with financial tools

Preferred devices: iPhone and laptop

Archetype: “The learner”

Personality: dreamer, curious, motivated, cautious

Narrative / Lifestyle / Behavioural patterns: Sofia has always been curious about investing but finds the terminology and complexity intimidating. She follows finance influencers and watches YouTube tutorials but hasn’t yet taken the leap into real investing.

She appreciates tools that make learning visual, interactive, and easy to understand. In her free time, she uses apps for personal growth, budgeting, and productivity

Objectives / Needs:

- Learn how to invest safely and progressively
- Experiment with different strategies without financial risk
- Understand the impact of her decisions through clear visual feedback
- Receive AI-driven guidance and insights

Frustrations / Pain points:

- Financial terminology and complex interfaces
- Lack of personalized feedback in most trading apps
- Difficulty staying motivated when results take time
- Uncertainty about trusting automated (AI) investment tools

Persona 2

Name: Pedro Soares

Quote: “I need tools that match my pace and let me stay in control.”

Age: 34 years

Education level: Master’s degree in Computer Engineering

Work/occupation: Freelance Software Developer

Family: Married, one child

Location: Lisbon, Portugal

Tech proficiency: High experience with data, algorithms, and digital tools

Preferred devices: Laptop, tablet, and smartphone

Archetype: “The strategist”

Personality: analytical, ambitious, competitive, tech-savvy

Narrative / Lifestyle / Behavioural patterns: Pedro has been trading stocks and cryptocurrencies for several years. He enjoys testing new strategies, following financial news, and optimizing performance. With a busy work and family life, he values automation and efficiency, preferring systems that can act on his behalf while still letting him adjust decisions.

He seeks tools that combine depth and flexibility, providing clear visualizations, quick insights, and opportunities to experiment with AI-driven strategies.

Objectives / Needs:

- Automate part of his trading process using AI
- Design and customize characters that reflect his investment style
- Compare different strategies and track their performance
- Receive real-time insights and feedback on results
- Save time without losing control over key decisions

Frustrations / Pain points:

- Too many disconnected trading platforms
- Slow or unclear feedback on performance
- Difficulty visualizing risk in an intuitive way
- Lack of transparency in automated (AI) decision-making

Activity scenarios

Activity scenario 1

Sofia has just returned home after work. She decides to open her new investment application that uses AI and characters with different personalities. After logging in, Sofia chooses the “Conservative” character, curious to see how it will manage her first simulation. She sets an initial amount of 100 euros and lets the AI start investing automatically. The app

displays real-time updates and small messages from the character explaining the decisions being made.

After a few weeks, Sofia checks her portfolio again and notices that it has grown by several euros. Excited with the results, she decides to sell part of her investments to lock in some profit. The app guides her through the selling process, showing how the transaction will affect her overall balance. She feels proud of her first success and starts planning to test an “Aggressive” character next time to explore a different strategy.

Activity scenario 2

After finishing a coding project, Pedro opens MyTradingPal to check on “AlphaVibe,” the custom AI character he created to trade automatically. He reviews the performance graphs and sees that AlphaVibe adjusted its portfolio after a recent market drop, keeping returns steady.

Curious to optimize further, Pedro tweaks a few parameters — lowering leverage and adding exposure to renewable energy stocks. The AI immediately simulates the new setup and shows improved long-term stability.

Satisfied with the results, Pedro lets AlphaVibe continue managing part of his portfolio. Encouraged by the app’s clear insights and automated performance, he plans to design a second, more aggressive character to compare strategies.

Functionalities

Core functionalities

Portfolio Monitoring:

- View real-time performance of assets, gains/losses, and balance
- Quick, glanceable dashboard with visual summaries and trend indicators
- Historical charts for performance tracking and comparisons

AI Character Interaction:

- Each AI persona represents a trading style (e.g., conservative, balanced, aggressive)
- Explainable AI: characters justify recommendations in plain language
- Real-time feedback as characters adjust strategies

Personalization and Customization:

- Customize character appearance, tone, and strategy behavior
- Set personal risk levels, investment goals, and learning objectives
- Adjust notification preferences and dashboard layout

Simulation and Learning Mode:

- Sandbox mode for risk-free strategy testing
- Educational guidance through AI commentary and tutorials
- Mode separation: clearly distinguishes simulation from real-money trading

Decision Management:

- Explicit confirmation dialogs for real-money actions
- Decision history with “why it worked/failed” insights

Supportive functionalities

Privacy and Security:

- Privacy mode to hide financial data in public settings
- Biometric login (Face ID/Fingerprint) and end-to-end encryption
- Automatic session timeout and secure logout

Offline Portfolio Access:

- View last-synced data when offline
- Automatic synchronization when connection is restored
- Data freshness indicator showing last update time

Explainable and Transparent AI:

- All recommendations include simple natural-language reasoning
- Optional “See details” for deeper insights into decisions
- Builds user trust and understanding of AI behavior

Accessibility and Usability:

- Scalable text and high-contrast themes
- One-handed mobile navigation and large touch targets

Notifications and Summaries:

- Smart daily summary notifications instead of constant alerts
- Customizable notification content and timing

Part II - Lo-fi prototype and heuristic evaluation

Project abridged description

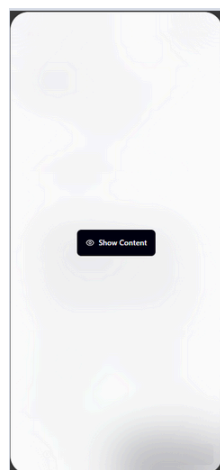
MyTradingPal is a mobile-first investing app that blends AI-driven portfolio management with interactive, character-based learning. Users can invest real money or practice in a simulation mode, guided by AI personas that clearly explain their decisions. The app offers real-time performance tracking, explainable recommendations, customizable risk levels, and strong privacy features like biometrics and public privacy modes. It's designed to make investing accessible and educational, helping users build confidence and financial literacy through a personalized AI experience.

These three tasks/features were implemented in the app's Lo-fi prototype using Figma:

- Task #1 - Check Portfolio Performance (Low Complexity);
- Task #2 - Configure & Launch AI Character (Medium-High Complexity)
- Task #3 - Analyze AI Recommendation & Execute Trade (High Complexity).

Prototype's Wireflow

Check Portfolio Performance



1st Step - We open the AI recommendation section in hidden mode



2nd Step - We can then switch to unhidden mode

3rd Step - Scroll down a bit until the portfolio's performance is visible

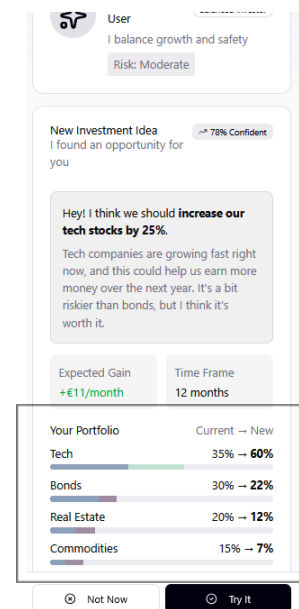
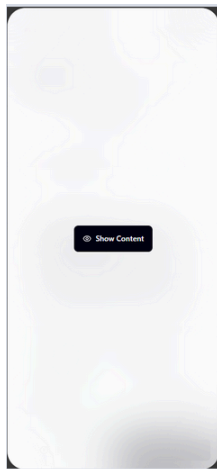


Image 1 - Task 1

Configure & Launch AI Character



1st Step - We open the AI recommendation section in hidden mode



2nd Step - We can then switch to unhidden mode

Step 3 - Scroll down until the end, click "Choose your persona" followed by the plus (+) sign

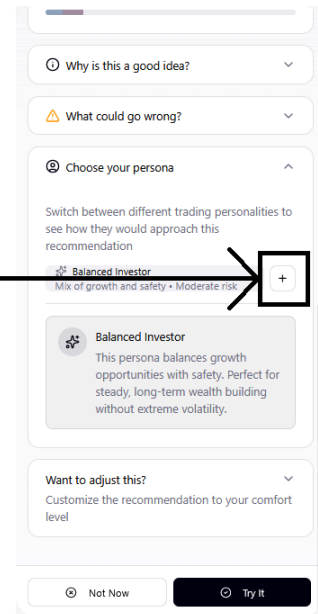
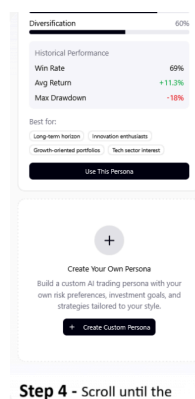
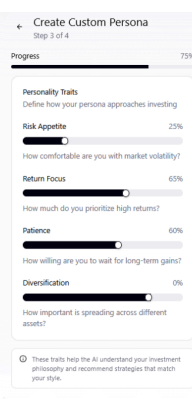
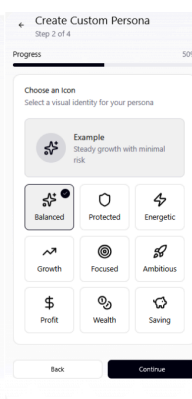
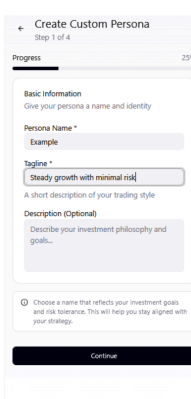


Image 2 - Task 2 Part 1



Step 4 - Scroll until the end again and click "Create Custom Persona"



Step 5 - Customize the persona as you see fit. After clicking "Save Persona", it is automatically selected.

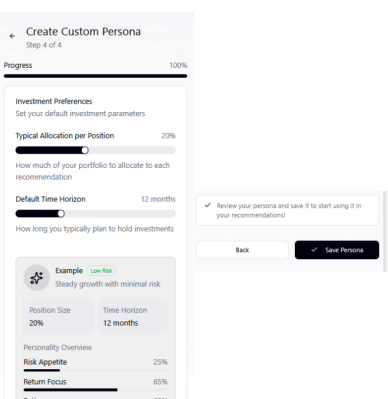
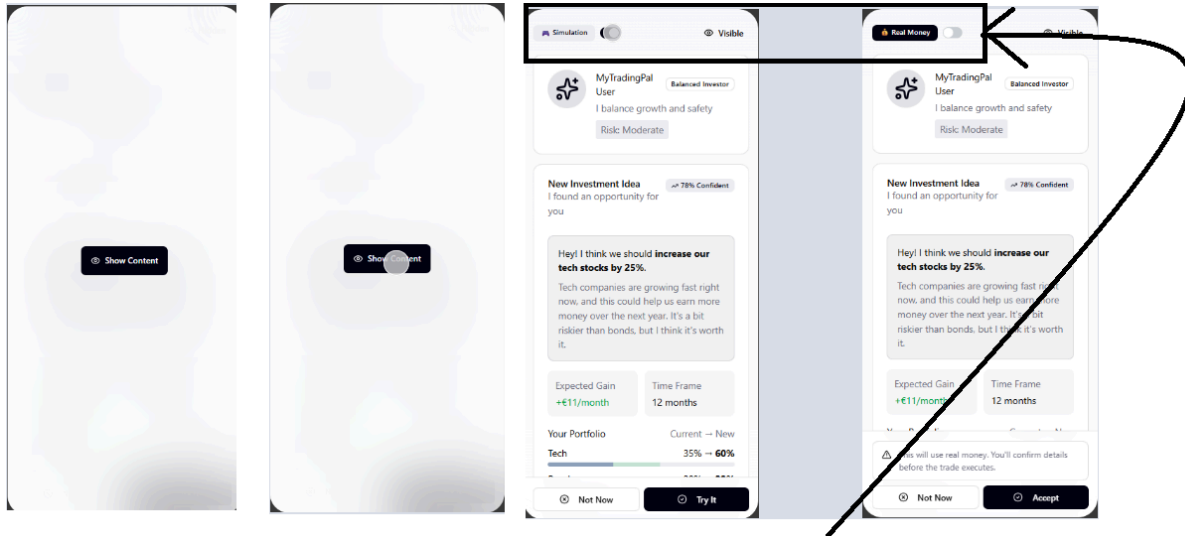


Image 3 - Task 2 Part 2

Analyze AI Recommendation & Execute Trade

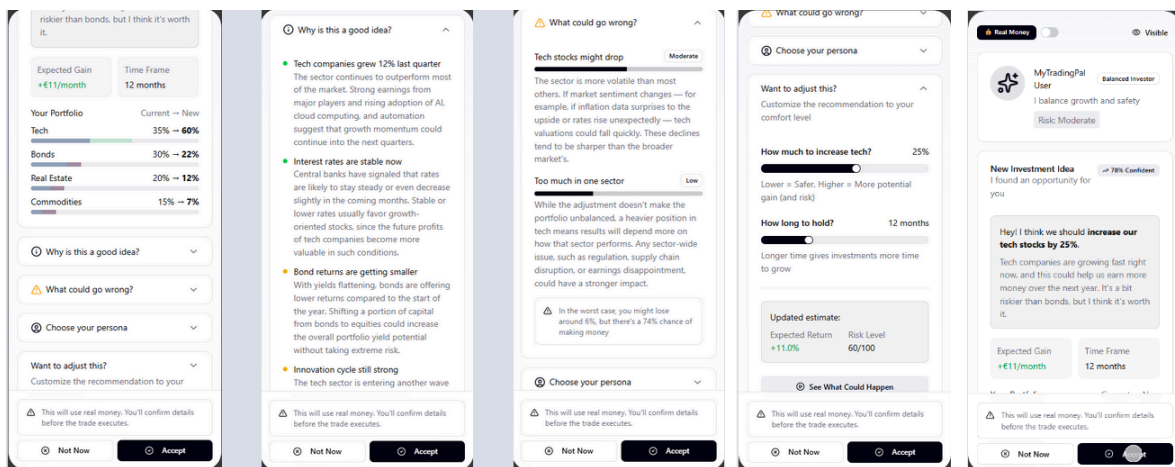


1st Step - We open the AI recommendation section in hidden mode

2nd Step - We can then switch to unhidden mode

3rd Step - After switching to not hidden mode, we can switch between simulation and real world

Image 4 - Task 3 Part 1



Step 4 - We can now analyze the reasons for the AI recommendation

Step 5 - We can also manually adjust the investment

Step 6 - We can finally either accept or reject the recommendation

Image 5 - Task 3 Part 2

Heuristic Evaluation Results

Problem #1 - Inefficient Interaction Flow

Identified by group 3, the current design forces the user into an inefficient interaction mode (switching identity) to achieve a simple information goal (comparison). The system makes

the user work to access a core value proposition: seeing how personalized the recommendation truly is. This violates Heuristic 7 - Flexibility And Efficiency of Use - and was classified with a severity of 3.

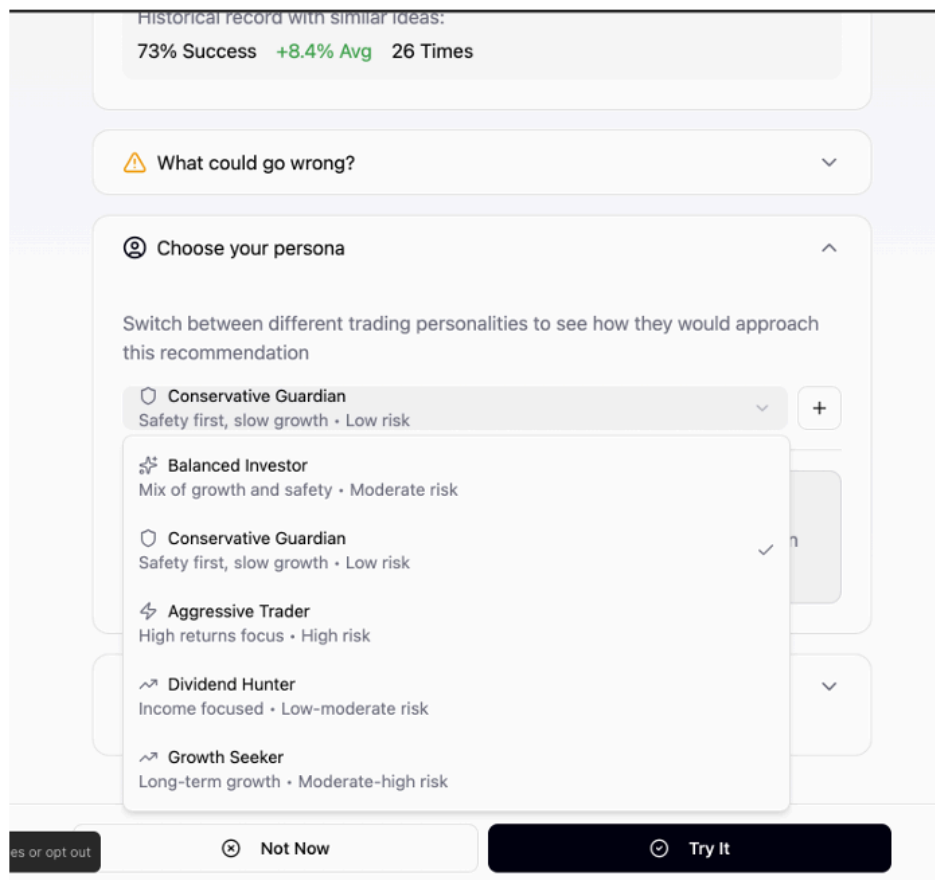


Image 6 - Inefficient Interaction Flow

As a group, we agree with this observation that the persona-switching flow is inefficient. Requiring users to change identities just to compare recommendations adds unnecessary friction, so we support their assessment.

Problem #2 - Lack of Risk Transparency

Identified by group 3, the specific risk factors (Moderate/Low bars) are presented without a clear link to the overall, quantitative Risk Level (60/100). The system lacks transparency on how the parts contribute to the whole risk score. This violates Heuristic 1 - Visibility of System Status - and was rated with a severity of 2.

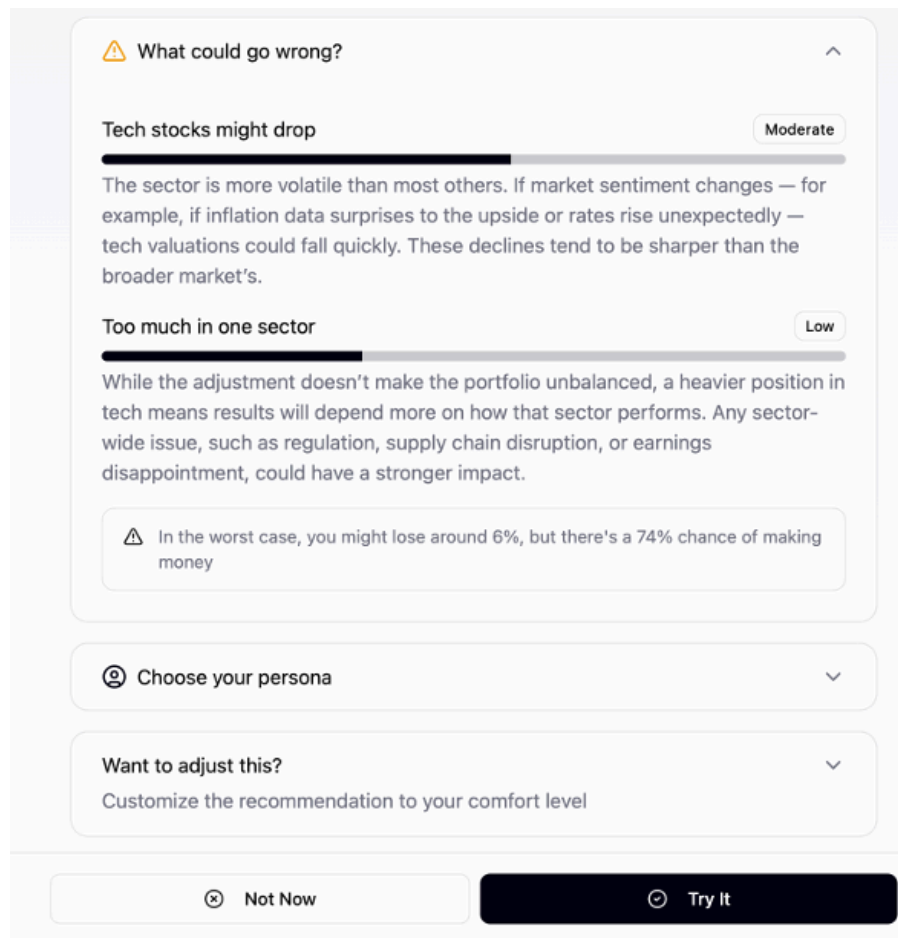


Image 7 - Lack of Risk Transparency

Our group is in line with group 3's identification of this as a significant usability issue. Their assessment effectively highlights a critical gap in the system's transparency regarding risk calculation and composition.

Problem #3 - Inefficient Persona Comparison

Identified by group 3, to compare the recommendation to another strategy, the user is forced to switch their active profile via a dropdown. This is a cumbersome, multi-step process that hinders efficient exploration and violates Heuristic 7 - Flexibility And Efficiency of Use - being graded with a severity of 3.

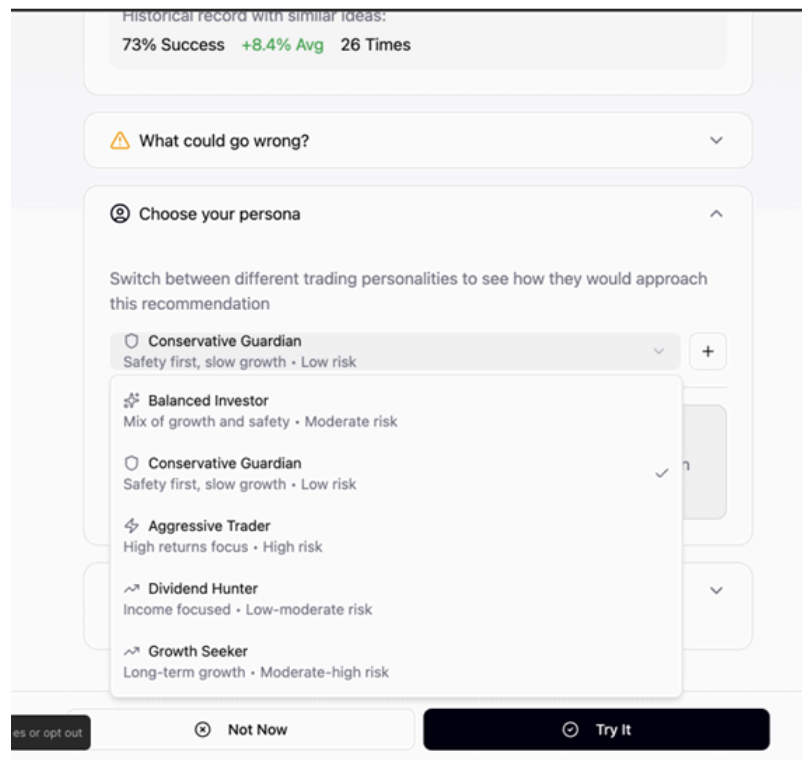


Image 8 - Inefficient Persona Comparison

Because this problem shares the same root cause as Problem #1, any solution that addresses the inefficient interaction flow would naturally resolve this issue as well. This makes addressing both problems together a logical and efficient approach to improving the system's usability.

Problem #4 - Missing Help and Documentation

Identified by group 4, the application lacks any help button, tutorials, or documentation section. Users have no way to consult when encountering questions about features or investment concepts, such as custom AI creation or risk level interpretation. This violates Heuristic 10 - Help and Documentation and was not classified with a degree of severity.

We agree this is a valid concern. Investment applications inherently involve complex financial concepts, making help resources essential rather than optional. The absence of any guidance mechanism leaves users stranded when questions arise, especially since our app is oriented towards beginners.

Corrections to perform in Phase 3

Solution #1 - Implement side-by-side comparison interface

Introduce a dedicated "Compare" function that enables users to view two AI persona cards side-by-side simultaneously. This solution directly addresses both Problem 1 and 3.

The comparison view would display key information for both personas in parallel, including the specific investment recommendations, persona description, risk level, position allocation, investment timeline, personality traits (Risk Appetite, Return Focus, Patience, Diversification), historical performance metrics (Win Rate, Average Return, Max Drawdown), and suitability tags.

By eliminating the need to switch between personas via the dropdown menu and mentally track differences, users can directly compare how different strategies approach their investment goals. This streamlines the decision-making process and resolves the core inefficiency underlying both identified problems.

Solution #2 - Enhance risk score transparency

Display the proportional contribution of each factor (in 'What could go wrong?') to the total score with expandable tooltips explaining the calculation methodology. This solution directly addresses Problem 2 – Lack of Risk Transparency.

The enhanced risk view would show each specific factor alongside its quantitative contribution to the overall Risk Level. Expandable tooltips would provide detailed explanations of the calculations behind each factor, illustrating how individual components combine to produce the total risk score.

By providing a clear mapping between individual risk factors and the overall score, users can understand the system's scoring logic without guessing or performing mental calculations.

Solution #3 - Develop comprehensive help system

Add a persistent "?" icon accessible from all pages, providing contextual assistance based on the user's current location within the application. This solution directly addresses Problem 4 – Missing Help and Documentation.

The enhanced help system would provide interactive tutorials for new users, a dedicated FAQ section covering common functionality and terminology questions, hover tooltips for technical or financial terms, and AI-powered assistance that allows users to ask questions in natural language and receive contextual guidance tailored to their current page or task.

By making guidance easily accessible, users can consult the system whenever questions arise about features, such as custom AI creation or investment concepts (like risk level interpretation), without leaving the app. The addition of AI help ensures users receive immediate, personalized explanations, improving understanding and reducing frustration.

Part III - Hi-fi prototype and user evaluation

Prototype's Wireflow

Check Portfolio Performance

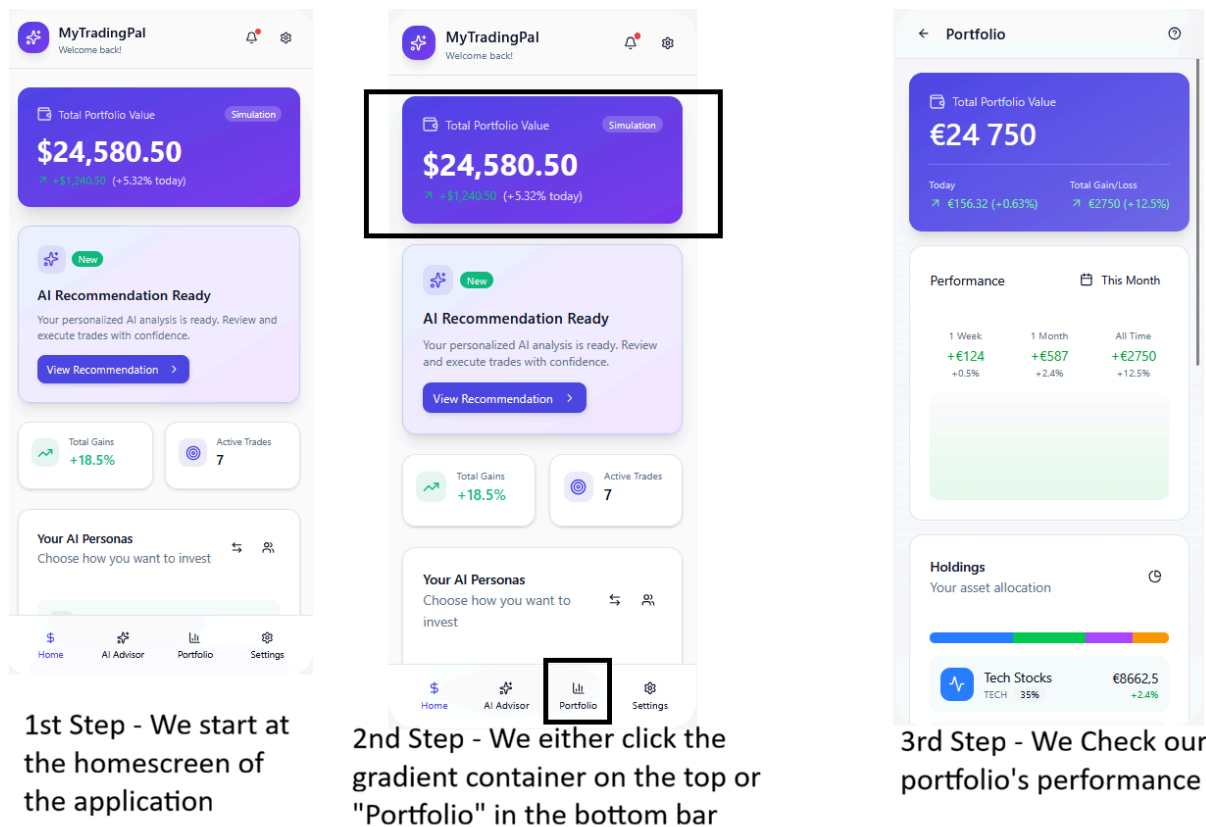
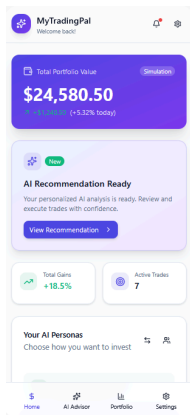
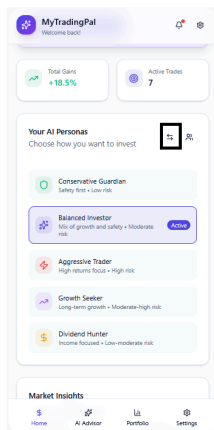


Image 9 - Task 1 Phase 3

Configure & Launch AI Character



1st Step - We start at the homescreen of the application



2nd Step - Scroll down until you see "Your AI Personas" and click the button with the two symmetric arrows

3rd Step - Select two personas to compare their configurations

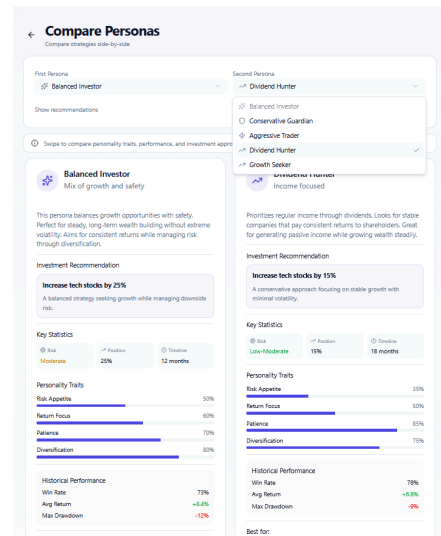
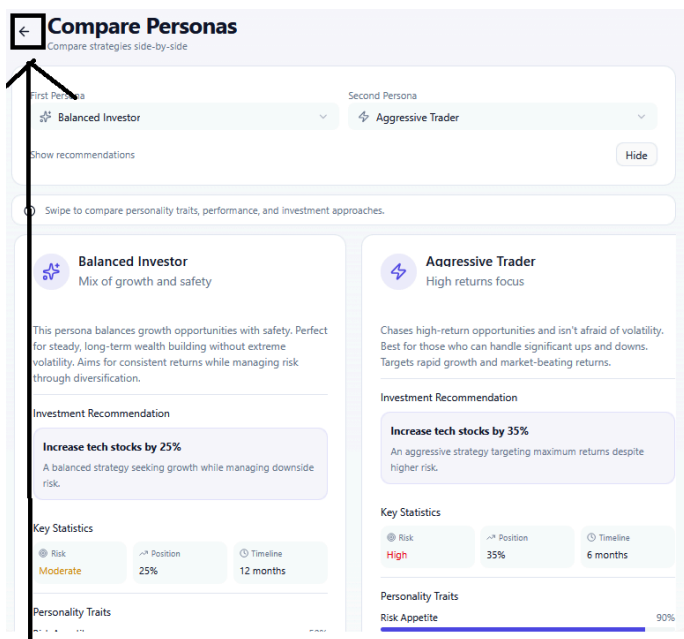
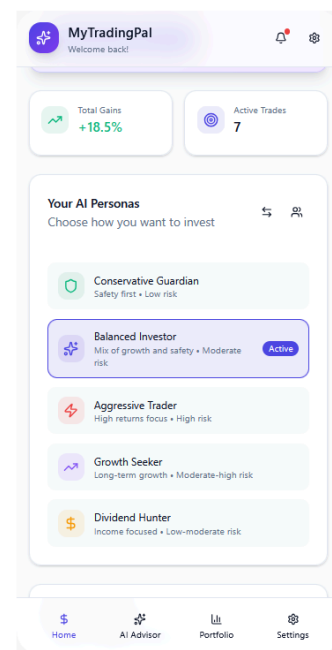


Image 10 - Task 2 Part 1 Phase 3



4th Step - After careful comparison, go back to the homescreen



5th Step - Scroll until you see the "Your AI Personas" section again and select the Persona you wish

Image 11 - Task 2 Part 2 Phase 3

Analyze AI Recommendation & Execute Trade

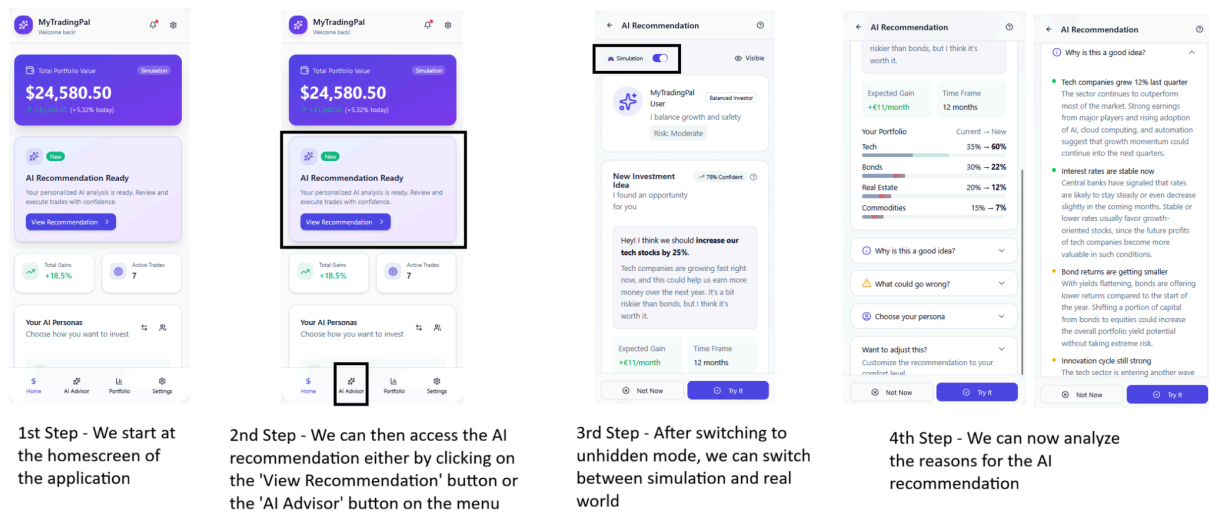


Image 12 - Task 3 Part 1 Phase 3

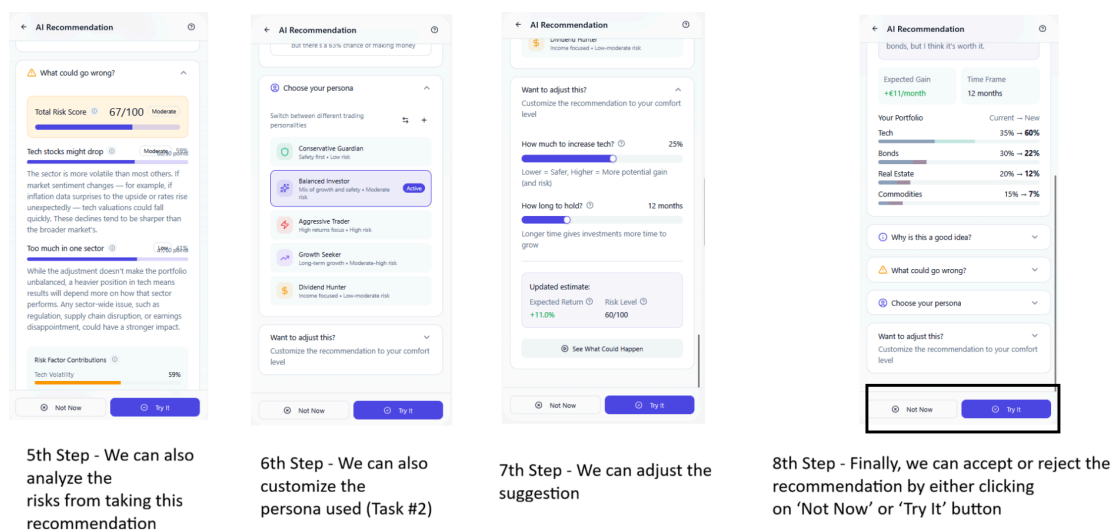


Image 13 - Task 3 Part 2 Phase 3

User evaluation protocol

The protocol detailed below describes the methodology used to test the MyTradingPal prototype with young adults (aged 18-30). While the following scenarios and scripts are presented in English to align with this report, the actual evaluation was carried out in Portuguese to accommodate the participants' native language. The original Portuguese protocol is in the Annexes section.

Objective

- Analyze the usability of the MyTradingPal high-fidelity prototype, specifically focusing on the effectiveness and efficiency of the new persona comparison, risk transparency, and help system features.
- The goal is to identify usability issues that prevent users from completing main tasks efficiently and effectively.

Target Users

- Target Profile: Young adults (18-30 years old), students or recent graduates with varied levels of financial literacy (beginners and intermediate).
- Recruitment: Convenience sampling (colleagues, friends) who fit the profile.
- Sample: At least 10 participants.

Methodology

1. Start with the Pre-Test and prepare the High-Fidelity Prototype.
2. Proceed with the introduction, explaining the prototype and what is intended to be evaluated.
3. Conduct the Pre-Test Questionnaire (described below) and allow the interviewee to read the application's introductory slides.
4. Begin Task Execution, recording the Measures for each Task.
5. At the end, conduct the Post-Test Questionnaire.

Pre-test

Preparation:

1. Load the High-Fidelity Prototype on the test device.
2. Explain to the interviewee that everything will be anonymous and only numbers will be collected.

Introduction:

- "Hello and thank you for participating. We are testing a new financial application called MyTradingPal, which uses AI characters to help manage investments".
- "Before we begin, I want to emphasize that we are testing the system, not you".
- "If you find something confusing or difficult, that is a problem with the interface, and not you".
- "Your honesty is essential to us".
- "During the tasks, I am going to ask you to think aloud and tell me what you are thinking, what you are looking for on the screen, and why you made certain decisions".

- "Do you have any questions?".

Pre-test Questionnaire

It does not need to be an actual questionnaire if the person is known, but note the following data:

- Age
- Investment Experience (beginner, intermediate, expert).

Measures

- During the Pre-Test, statistical data regarding the age group being tested was noted: Age, and if they had ever invested before.
- During the execution of each task, the following quantitative and qualitative data were noted and recorded:
 - Time taken to execute the task.
 - Number of errors (misclicks) made.
 - Whether the interviewee completed the task.
 - Whether the interviewee requested assistance during the task.
 - How the interviewee classified the task on a scale of 1 to 5 (Very Difficult to Very Easy).
 - What confused the interviewee (short answer).

Task Execution

Task 1:

- Scenario: Imagine that you are an investor in this application. Please find a way to view your portfolio.

Task 2:

- Scenario: You received a new investment recommendation. You are undecided whether you should use a Balanced strategy or an Aggressive one.
- Please use the Compare functionality to see the two personas side-by-side.
- After analyzing the differences, select the Balanced Investor.

Task 3:

- Scenarios: You selected the Balanced Investor. You see a recommendation to increase Technology stocks, but there is an associated Risk Level.
 1. Try to discover exactly why the risk level has that value (look for details about the factors).

2. If you do not understand a financial term or functionality on this page, try to find Help within the application.
3. When you are satisfied, accept the recommendation.

Post-test Questionnaire

Fill out the System Usability Scale (SUS) (Scale of 1 to 5, Disagree - Agree):

- Would you like to use this application frequently?
- Did you find the system unnecessarily complex?
- Did you need technical help to use this system?
- Are the system functions well integrated?
- Was there inconsistency in this system?
- Did you think that the majority of people (between 18 and 30) would learn to use this system quickly?
- Did you feel confident using this system?

Conclusion

This concludes our evaluation. Thank you very much for your time. Your feedback will be essential for us to improve MyTradingPal.

Results

Sample characterization

A total of 16 participants took part in the usability evaluation, exceeding the minimum sample size of 10 outlined in the protocol. The sample consisted of young adults aged between 18 and 23 years ($M = 20.44$, $SD = 1.63$), fully aligning with the target profile of students and recent graduates (18-30 years).

Regarding financial literacy and investment experience, the group was evenly balanced to ensure the system catered to both user types:

- 8 participants (50%) reported having prior investment experience (Intermediate/Experienced).
- 8 participants (50%) stated they had never invested before (Beginners).

This split allowed for a comprehensive analysis of the interface, testing whether MyTradingPal effectively supports novice users while remaining efficient for those with existing financial knowledge.

Results and statistical analysis of task 1

This task aimed to validate the visibility of the system status and the clarity of the dashboard.

Pessoa Nº	Tempo	# Erros	Concluiu?	Assistência?	Como classificarias esta tarefa?	O que foi mais confuso nesta tarefa?
1	0:07:00	1	Sim	Sim	5	Nada
2	0:21:00	2	Sim	Não	3	Não
3	0:16:00	1	Sim	Não	5	
4	0:08:00	1	Sim	Não	5	
5	0:34:00	3	Sim	Não	4	
6	0:17:00	1	Sim	Não	5	Nada
7	0:14:00	1	Sim	Não	5	Nada
8	0:31:00	1	Sim	Não	4	
9	00:10:00	1	Sim	Não	5	
10	0:09:00	1	Sim	Não	5	
11	00:22:00	1	Sim	Não	4	
12	00:29:00	1	Sim	Não	5	
13	00:27:00	1	Sim	Não	5	
14	00:18:00	2	Sim	Não	5	Fácil
15	00:12:00	2	Sim	Não	5	
16	00:36:00	1	Sim	Não	4	

Image 14 - Task 1 Results

Quantitative Analysis:

- Success Rate: 100%. All 16 participants successfully identified their portfolio performance without critical failures.
- Time on Task: The average completion time was 19.3 seconds. The 95% Confidence Interval for this task is [14.3s, 24.3s], indicating a high degree of certainty in the efficiency of the interface for this specific action. Outliers in the data (users taking >30s) were attributed to external distractions rather than UI obstacles.
- Error Rate: Participants committed an average of 1.3 errors per user. While slightly higher than predicted for a low-complexity task, this remains within acceptable limits for a first-time interaction.
- Perceived Difficulty: The task was rated 4.6/5.0 (where 5 is Easiest), confirming that the portfolio dashboard is intuitive and requires minimal cognitive load.

Qualitative Analysis:

- Users described the task as simple, with comments such as "Fácil" indicating a smooth flow.
- The high success rate validates the visibility of the "Total Portfolio Value".

Results and statistical analysis of task 2

This task tested the user's ability to customize the AI persona, a core differentiator of the MyTradingPal app.

Pessoa Nº	Tempo	# Erros	Concluiu?	Assistência?	Como classificarias esta tarefa?	O que foi mais confuso nesta tarefa?
1	1:03:00	3	Sim	Sim	3	O símbolo das duas setas causa confusão e é confundido com transferência
2	2:20:00	4	Sim	Sim	3	O símbolo representa bem, mas encontra-se muito escondido, tem que estar mais saliente
3	0:10:00	0	Sim	Sim	5	É preciso maior detalhe no símbolo
4	0:08:00	0	Sim	Não	5	O ícone está pouco salientado e não explica bem
5	0:16:00	1	Sim	Não	4	O ícone está confuso, muito escondido
6	01:11:00	2	Sim	Não	4	Botão de acesso não claro
7	0:35:00	0	Sim	Não	4	Nada
8	2:01:00	5	Sim	Sim	3	Ícones muito semelhantes
9	0:42:00	0	Sim	Não	5	
10	1:38:00	3	Sim	Sim	4	O botão podia ter rótulo
11	00:51:00	7	Sim	Sim	3	não haver um sítio direto para fazer a comparação
12	00:41:00	3	Sim	Não	3	O ícone não faz sentido
13	00:37	2	Sim	Não	5	
14	00:20:00	0	Sim	Não	5	Fácil
15	00:46:00	2	Sim	Não	4	
16	0:45:00	3	Sim	Não	5	O ícone de duas setas confunde com transferências

Image 15 - Task 2 Results

Quantitative Analysis:

- Success Rate: 100%, although 20% (3 participants) required some form of assistance to complete the flow.
- Time on Task: The average completion time was 52.8 seconds. The 95% Confidence Interval for this task is [32.4.s, 73.1s]. The significantly wider confidence interval compared to Task 1 highlights a disparity in user experience: while some completed it quickly, others struggled significantly.
- Error Rate: Participants committed an average of 2.2 errors per user. This higher error rate correlates directly with the ambiguity of specific icons in the interface.
- Perceived Difficulty: 4.1 / 5.0. Although statistically rated as "Easy," qualitative feedback reveals this score is misleading; users found the task easy after receiving assistance, indicating a discoverability issue rather than a complexity issue.

Qualitative Analysis:

- Icon Ambiguity: The primary friction point was the "Two Arrows" icon used for comparing personas. Qualitative feedback explicitly stated: "O símbolo das duas setas causa confusão e é confundido com transferência" and "O ícone de duas setas confunde com transferências".
- Approximately 27% of users misinterpreted this icon, leading to hesitation and errors.
- Replacing the abstract "Two Arrows" icon with a clearer metaphor (for example a scale or text label "Compare") is expected to resolve this bottleneck.

Results and statistical analysis of task 3

The final task evaluated the complete investment loop: analyzing an AI suggestion, adjusting risk parameters, and executing a trade.

Pessoa Nº	Tempo	# Erros	Concluiu?	Assistência?	Como classificarias esta tarefa?	O que foi mais confuso nesta tarefa?
1	1:25:00	8	Sim	Sim	4	Os menus colapsados por default causam confusão
2	4:41:00	9	Sim	Não	5	Nota: utilizador passou tempo a aprender a aplicação
3	0:50:00	2	Sim	Não	5	O sub menu de ajuste não está muito bem sinalizado
4	0:17:00	0	Sim	Não	5	O sub menu está pouco destacado
5	0:16:00	0	Sim	Não	5	O sub menu podia ter ícone ou já estar aberto
6	2:05:00	5	Sim	Sim	4	Falta de ícone
7	0:58:00	1	Sim	Não	5	O menu podia ser mais notável
8	3:22:00	7	Sim	Sim	3	Excesso de informação no ecrã inicial
9	1:10:00	1	Sim	Não	5	
10	2:47:00	6	Sim	Sim	4	Não estava claro o submenu
11	01:18:00	8	Sim	Não	3	
12	00:50:00	5	Sim	Não	4	O menu podia estar mais claro
13	01:07:00	6	Sim	Sim	3	Muita informação para começar sem um "tutorial"
14	00:46:00	3	Sim	Não	4	Pouco destacado
15	01:03:00	5	Sim	Não	4	
16			Não		2	

Image 16 - Task 3 Results

Quantitative Analysis:

- Success Rate: 93.8% (15/16 participants) completed the task.
- Time on Task: The average completion time was 91.7 seconds. The 95% Confidence Interval for this task is [51.2s, 132.2s]. This extremely wide interval confirms the "Usability Paradox" observed in the study: while the task is logically simple, the discovery of the necessary controls varies wildly between users.
- Error Rate: The average number of errors was 4.4 per user. This high count was driven by visibility issues rather than a lack of understanding.
- Perceived Difficulty (The Usability Paradox): Despite the high error rate, users rated the task 4.1/5.0 in ease of use. This suggests users understood what to do but struggled to find where to do it.

Qualitative Analysis:

- Hidden Controls: The primary source of errors was the "Adjust" menu being collapsed by default. Users noted: "Os menus colapsados por default causam confusão" and "O sub menu de ajuste não está muito bem sinalizado".
- Discovery vs Complexity: Feedback confirms the issue was discoverability, not complexity. Users explicitly mentioned that once found, the logic was clear.
- Expanding the adjustment controls by default will likely eliminate the majority of errors without increasing the actual complexity of the task.

SUS Scores

To understand how users truly felt about the app, we asked them to fill out a standard "System Usability Scale" questionnaire after finishing the tasks.

Pessoa Nº	Gostarias de usar esta aplicação com frequência?	Achaste o sistema desnecessariamente complexo?	Precisavas de ajuda técnica para usar este sistema?	As funções do sistema estão bem integradas?
1	4	2	1	4
2	4	1	2	3
3	4	2	1	4
4	3	2	2	4
5	4	1	1	4
6	4	2	1	4
7	5	1	1	5
8	3	3	2	3
9	5	1	1	5
10	4	2	1	4
11	5	2	3	5
12	5	3	2	4
13	4	2	2	4
14	4	2	2	5
15	4	2	2	4
16	4	2	2	4

Image 17 - SUS Scores Part 1

Pessoa Nº	Havia inconsistência neste sistema?	Achaste que a maioria das pessoas (entre os 18 a	Sentiste-te confiante ao usar este sistema?
1	1	3	4
2	4	5	5
3	4	5	5
4	3	4	4
5	2	5	5
6	2	4	5
7	1	5	5
8	3	3	3
9	1	5	5
10	2	4	5
11	2	5	3
12	2	4	4
13	1	4	5
14	3	3	4
15	1	5	4
16	2	4	4

Image 18 - SUS Scores Part 2

The results told an interesting story: users really liked the app's logic, even if they got stuck on a few hidden buttons.

Overall, the scores were very positive. When asked if they would use MyTradingPal frequently, the vast majority gave high ratings. This is a great sign. It means that once users get past the initial issue of learning what the icons mean, they see real value in the tool.

Interestingly, we found a "usability paradox." Users struggled to find the adjustment menu in the final task, yet they still rated the system as "Easy" to use and not complex. Most participants rated the complexity as very low (1 or 2 out of 5 for the question "Achaste o sistema desnecessariamente complexo?") and reported feeling confident while using it. This suggests they knew exactly what they wanted to do - they just couldn't find where to click. They didn't feel confused by the finance concepts, just annoyed by a hidden menu.

Participants also agreed that the system's functions were well integrated. The transition from checking their portfolio to getting advice from the AI felt smooth and logical to them.

It is worth noting that our testers were mostly university students who are good with technology. While this might make the scores slightly higher than average, the fact that even tech-savvy users got stuck on specific icons validates that those visual bugs are real problems.

Discussion

Our global analysis reveals a very clear story about MyTradingPal: the "engine" of the application is powerful and logical, but the visual interface needs some specific repairs to match that quality.

On the bright side, the fundamental design proved to be highly effective. An impressive 94% of participants successfully completed every single task assigned to them, and satisfaction ratings remained consistently high (above 4 out of 5) even when users hit small roadblocks. This is a critical win because it proves that the product's utility outweighs the learning curve: users are willing to forgive small design flaws because they see value in the tool. The positive System Usability Scale scores confirm that users trust the core logic and flow of the system. However, we must be mindful that these results might be slightly optimistic. Our testing pool consisted of university students and recent graduates who are not only naturally more comfortable with technology than the average person but were also people close to us. This reliance on convenience sampling means our feedback may be friendlier or more forgiving than what we might encounter in a broader market test .

The primary bottlenecks we identified were never about how to perform a task, but simply finding the right button to start it. For instance, the "Two Arrows" icon confused over a quarter of our users, and the collapsed adjustment menus tripped up one third of the participants. The impact of these small visual "hiding spots" was massive: users who missed these specific cues took up to three times longer to finish a task and committed three times as many errors as the average user.

In conclusion, the application is functionally sound, and visually, it only requires targeted fixes rather than a complete overhaul. We are confident that by correcting just three specific visual elements (the ambiguous comparison icon, the hidden default menus, and the small help button) we can eliminate the vast majority of user errors. Future testing should focus on validating these specific fixes with a wider, less tech-savvy audience to ensure the app is truly ready for everyone.

Conclusion

The primary objective of Phase 3 was to validate the high-fidelity prototype of MyTradingPal through rigorous user testing, confirming whether our design decisions successfully addressed the "financial anxiety" identified in Phase 1. The results provide strong evidence that we have achieved a solid foundation: the application's core logic is robust, the "AI Mentor" concept resonates with our target audience, and the integration between portfolio management and education is seamless.

From a broader project perspective, MyTradingPal has successfully evolved from a concept addressing financial illiteracy into a validated prototype. We moved from identifying user personas like "Sofia" (the anxious beginner) to building a solution that users explicitly stated they would "use frequently". The "Usability Paradox" we observed (where users rated the app as easy despite making errors) indicates that we have solved the hard problem of making finance approachable, and now only need to solve the easier problem of making buttons more visible.

Moving forward, the immediate next steps are clear: expanding the "Adjust" menus by default, redesigning the "Compare" icon to be more literal, and increasing the visibility of the "Help" system. By addressing these specific discoverability bottlenecks, we can refine MyTradingPal from a promising prototype into a polished, market-ready tool that truly empowers young adults to invest with confidence.

Annexes

Phase 1 Annexes

Questionnaire

- <https://forms.gle/CgBJgayGpRhUpBQP6>

Results

- <https://docs.google.com/document/d/1CPXZpOW-sKNJAcUpiZMqwdZBekDM-GQQMAq6Y71kKiM/edit?usp=sharing>

Phase 2 Annexes

Sent Heuristic Evaluation Reports

- Group 1:  ipc-heuristiceval-T03-G1-G5.pdf
- Group 6:  ipc-heuristiceval-T03-G6-G5.pdf

Received Heuristic Evaluation Reports

- Group 3:  ipc-heuristiceval-T03-G05-G03.pdf
- Group 4:  ipc-heuristiceval-T3G5-G4.pdf

Phase 3 Annexes

Evaluation Protocol

-  EvaluationProtocol.docx

Evaluation Results

-  IPC : Entrevistas (Respostas)

Prototype

- <https://sepia-ratio-43320836.figma.site/>